

Dawson Kinsman

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🐙 github.com/dkinsman

Education

Michigan State University

2024–Present

PhD in Computational Mathematics, Science and Engineering (CMSE) (Advisor: Danny Caballero)

East Lansing, MI

- **Fellowships:** College of Engineering Distinguished Scholar Fellow, AI and Data enabled predictive Multiscale Modeling across STEM NSF Research Traineeship (AIDMM-NRT) Fellow
- **Certificates:** The Erdős Institute Data Science Boot Camp

University of Michigan-Dearborn

GPA: 4.00/4.00

Master of Science in Applied and Computational Mathematics (Advisor: Thomas Fiore)

Dearborn, MI

- **Awards:** Departmental Award for Excellence: Applied and Computational Mathematics

University of Michigan-Dearborn

GPA: 3.99/4.00

Bachelor of Art in Mathematics, Applied Statistics

Dearborn, MI

- **Certificates:** Practical Aspects of Computer Security
- **Awards:** Carl Rasmussen Award for Excellence: Applied Mathematics (2 years), Dean's List (8 semesters)

Research

Graduate Research Assistant

Aug 2024 – Present

Michigan State University

East Lansing, MI

- Trained and tested Latent Dirichlet Allocation (LDA) models for cluster analysis of education research abstracts. Contributed to construction of data cleaning and clustering pipeline for abstracts and new test abstracts.
- Prepared data application for Michigan Education Data Center to conduct a data mining and modeling project evaluating effects of STEM post-secondary training readiness and attainment.

Volunteer Researcher

Jan 2023 – Present

Computational Epidemiology Dispersed Volunteer Research Network

Remote

- Created web-scraping tools in Python to systematically obtain Google search trends data related to ShotSpotter and policing.
- Collaborate with external researchers on ShotSpotter analysis in Detroit (discussed further below).

Student Research Assistant

Jan 2022 – Present

University of Michigan – Dearborn

Dearborn, MI

- Implemented topological data analysis (TDA) methods to analyze police shooting data. Prepared and revised TDA findings in a co-authored report submitted to PLOS One.
- Led statistical analysis of Detroit Police Department 911 calls open data to examine the effect of a gunshot acoustic-detection system (ShotSpotter) on policing and crime metrics for future policy considerations. Contributed to a public dashboard containing general ShotSpotter findings for the broader Detroit community.
- Implemented a new empirical Bayes method in R for a binary classification problem with a small sample size and high dimensional data. Prepared and revised manuscripts for submission to MDPI Genes.

Undergraduate Research Assistant

Jun 2022 – Jul 2022

Big Data Summer Institute—University of Michigan

Ann Arbor, MI

- Researched topics in statistics, data science, and machine learning topics with a focus on developing analysis and R and Python skills.
- Coded in Python to data mine and investigate correlation between the rise of STIs, COVID-19, and other covariates.
- Collaborated on presentation to communicate statistical findings to a general audience.

Peer-Reviewed Publications

† equal contribution, * undergraduate author

- **Dawson Kinsman**, Zhi Zhang, Jian Hu, Gengxin Li. “New empirical Bayes models to analyze RNA-seq data from two different regions in hypophosphatasia disease study,” MDPI Genes, 2024. <https://doi.org/10.3390/genes15040407>.
- **Dawson Kinsman**^{†*} and Tian An Wong[†]. “Proactive Policing as Reinforcement Learning,” International Conference on Learning Representations (ICLR) Tiny Papers, 2023. Open Review.

In Revision

- **Dawson Kinsman*** and Tian An Wong. “The Homological Persistence of Police Violence: Analysis and Limitations,” PLOS ONE, 2024.

In Preparation

- **Dawson Kinsman**, Hadi Chaaban*, Divya Ramjee, Maimuna Majumder, Antonios Koumpias, and Tian An Wong. “Causal analysis of automated acoustic gunshot detection technology: Evidence from Detroit.” Draft.

Workshops

Reality Check: Spotting AI Shenanigans

Blurred Realities, MSU Broad Art Museum

Jan 30, 2026

East Lansing, MI

- Collaborate with other graduate students to develop 1 of the 16 projects selected from over 240 submissions to lead a workshop during the Blurred Realities exhibition.
- Conducted a hands-on workshop for undergraduates to introduce skills to help them identify misinformation and understand appropriate AI usage in information sharing. Provided practical strategies to assess AI capabilities, recognize bias and misleading content, and use data evidence to verify or debunk claims in an increasingly AI-driven world.

Dartmouth Scholar's Program

Dartmouth College

Oct 12, 2023 – Oct 15, 2023

Hanover, NH

- Presented a poster to general STEM audience and conducted mock graduate admissions interviews.
- Learned about ongoing research at Dartmouth and admissions to graduate programs at Dartmouth.

Service

Graduate Women in STEM (GWIS)

Secretary

May 2025 – Present

Computational Mathematics, Science, and Education Committees

1855 Computational Education Tenure-track Search

Feb 2025 – Present

Advisory Committee

Aug 2025 – Present

Computational Mathematics, Science, and Engineering Graduate Student Organization (GSO)

Vice President

Aug 2025 – Present

Secretary

Nov 2024 – Jul 2025

Mentorship

Undergraduate

August 2025 – Present

Graduate Women in STEM (GWIS)

East Lansing, MI

- Mentored 2 undergraduate students on applying for graduate school, finding research and career opportunities.

Work Experience

Data Science Intern

Jan 2024 – Aug 2024

Traxen

Plymouth, MI

- Led development and implementation of internal diagnostics to identify failing units and track their repairs. Automated the generation and dissemination of Excel reports to relevant teams.
- Contributed to constructing a new data processing pipeline reducing processing time by 30% while increasing the number of features generated.
- Create internal Streamlit dashboard connected to MongoDB database for daily diagnostics and pilot performance updates. Optimized NoSQL queries to reduce page load times and improve overall performance.
- Analyzed data to investigate correlations between fuel efficiency, driving conditions, and habits, and design features to best utilize these relationships in machine learning algorithms.

Strength in Numbers Mentee

Oct 2023 – Jan 2024

Detroit Lions Analytics Mentorship Program – Detroit Lions

Detroit, MI

- Explored the field of sports analytics. Collaborated on a confidential analytics project for the Detroit Lions.

Writing Center Consultant

Sept 2020 – May 2023

University of Michigan–Dearborn

Dearborn, MI

- Tutored students across multiple disciplines to improve writing and communication skills.
- Collaborated with other consultants to learn and teach current pedagogy and interdisciplinary writing.

Technical Skills

Languages: Python (Advanced), R (Advanced), C++ (Intermediate)

Technologies: PySpark, Pandas, NumPy, TensorFlow, $\text{\LaTeX}$, MongoDB.

Productivity: Notion, Jira, Microsoft 365, Google Workspace.